

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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CA-1 — ANSWER KEY

Class : V

Max Marks : 50

Subject : Mathematics

Duration : 1 Hr 30 Min

Syllabus: We the Travellers–I • Fractions • Angles as Turns

General Instructions: All questions are compulsory. Read each question carefully before answering.

Section A — 1 Mark Each

1. Which number has the same nearest ten and nearest hundred? (a) 7,126 (b) 7,835 (c) 6,999 (d) 7,030

Ans: (c) 6,999 (nearest ten = 7,000, nearest hundred = 7,000)

2. If $\frac{1}{3} = \frac{2}{x}$, find the value of x.

Ans: $x = 6$

3. Which fraction is the greatest: $\frac{5}{9}$, $\frac{5}{7}$, or $\frac{5}{12}$?

Ans: $\frac{5}{7}$

4. Starting from South, you make a full turn clockwise. Which direction do you end up facing?

Ans: South

5. 3 right angles equal how many quarter turns?

Ans: 3

6. Find the value: $10,000 + 900 + 20 + 4$

Ans: 10,924

7. True or False: An angle can be both acute and a full turn at the same time.

Ans: False

8. How many zeros are there in the number 60,000?

Ans: 4

Section B — 2 Marks Each

9. Choose the digits 1 and 9. Form two 2-digit numbers and find the result of the first subtraction only.

Ans: $91 - 19 = 72$

10. Explain in one line why the repeated-subtraction digit puzzle always ends at 9, no matter which two different digits are chosen.

Ans: Because the difference between any two-digit number and its digit-reversal is always a multiple of 9

11. For the digits 2 and 8, find (difference of digits) $\times$ 9, and check it against $82 - 28$.

Ans: $(8-2) \times 9 = 54$; $82 - 28 = 54$, so they match

12. Compare $\frac{3}{9}$ and $\frac{4}{8}$ using two methods: (i) simplifying each fraction (ii) comparing with $\frac{1}{2}$.

Ans: $\frac{3}{9} = \frac{1}{3} < \frac{1}{2}$; $\frac{4}{8} = \frac{1}{2}$; so $\frac{4}{8} > \frac{3}{9}$ by both methods

13. Give one example of a number whose nearest hundred and nearest thousand are the same, and explain why this happens.

Ans: e.g. 7,030: nearest hundred = 7,000, nearest thousand = 7,000; this happens because the tens/ones part is small enough that both round down to the same value

14. If the minute hand turns $\frac{4}{12}$ of a full turn from 12, how many minutes does this represent, and what is $\frac{4}{12}$ in its simplest form?

Ans: $\frac{4}{12} \times 60 = 20$ minutes; simplest form of $\frac{4}{12}$ is $\frac{1}{3}$

15. Starting from West: turn 3 right angles clockwise, then $\frac{1}{2}$ right angle clockwise, then another $\frac{1}{2}$ right angle clockwise. Find the final direction.

Ans: 3 right angles + $\frac{1}{2}$ + $\frac{1}{2}$ right angle = 4 right angles = 1 full turn; you end up facing West again

16. Find the fraction equivalent to $\frac{5}{9}$ with numerator 25, then compare it with $\frac{25}{54}$.

Ans: $\frac{5}{9} = \frac{25}{45}$; since $\frac{25}{45}$ and $\frac{25}{54}$ have the same numerator, $\frac{25}{45} > \frac{25}{54}$ (smaller denominator), so $\frac{5}{9} > \frac{25}{54}$

17. Ravi's car has been driven 56,987 km and Sheetal's car 67,543 km. Round each distance to the nearest ten-thousand and compare using only the rounded values.

Ans: 56,987 rounds to 60,000; 67,543 rounds to 70,000; by the rounded values, Sheetal's car has been driven more

18. Using the digits 4, 8, 2, 4, 7 (each used exactly once, as given), write the smallest and the largest possible 5-digit numbers.

Ans: Smallest = 24,478; Largest = 87,442

Section C — 3 Marks Each

19. A cyclist travels at 15 km/hr and a motorbike at 50 km/hr. In 3 hours, how much farther does the motorbike travel than the cyclist? Show your working.

Ans: Cyclist: $15 \times 3 = 45$ km; Motorbike: $50 \times 3 = 150$ km; Difference = $150 - 45 = 105$ km

20. Rearrange the digits of 48,247 (using all 5 digits) to form the smallest possible 5-digit number and the largest possible 5-digit number. State both.

Ans: Smallest = 24,478; Largest = 87,442

21. A school has 461 girls and 439 boys. If the school uses exactly 24 Minibuses (25 seats each) first, how many students remain, and how many Boats (20 seats each) are needed to carry them with no seats empty?

Ans: Total = 900; Minibuses carry $24 \times 25 = 600$; Remaining = $900 - 600 = 300$ students; Boats needed = $300 / 20 = 15$ boats exactly

22. Compare $\frac{12}{9}$ and $\frac{4}{5}$ using 1 (a whole) as a reference. If these represent pizzas eaten, which represents 'more than one full pizza'?

Ans: $\frac{12}{9} > 1$ (more than a whole) and $\frac{4}{5} < 1$; so $\frac{12}{9}$ represents more than a full pizza

Section D — 4 Marks Each (Word / Application Problems)

23. Town populations: Town 1 = 65,232; Town 3 = 56,380; Town 5 = 45,858; Town 6 = 66,540. (a) Arrange in descending order. (b) Round each to the nearest thousand. (c) State which towns' populations round up and which round down, with reasons.

Ans: (a) $66,540 > 65,232 > 56,380 > 45,858$. (b) 67,000; 65,000; 56,000; 46,000. (c) Town 1 (65,232) rounds down since hundreds digit $2 < 5$; Town 3 (56,380) rounds down since hundreds digit $3 < 5$; Town 5 (45,858) rounds up since hundreds digit $8 \geq 5$; Town 6 (66,540) rounds up since hundreds digit $5 \geq 5$

Section E — 2 Marks Each (From Your Textbook)

24. Take the digits 3 and 7. Form the two possible 2-digit numbers, subtract the smaller from the bigger, and repeat the process with the digits of each new result until you reach a 1-digit number. Show all your steps.

Ans: $73 - 37 = 36$; $63 - 36 = 27$; $72 - 27 = 45$; $54 - 45 = 9$. Final digit = 9

25. Rearrange the digits of 48,247 (using all 5 digits) to form the largest possible 5-digit number.

Ans: 87,442

26. Give one valid combination of exactly 7 fraction-kit pieces (from halves, thirds, quarters, fifths, sixths, eighths, or twelfths) that add up to a whole, and show your working.

Ans: 1 piece of $\frac{1}{2}$ + 6 pieces of $\frac{1}{12}$ = $\frac{1}{2} + \frac{6}{12} = \frac{1}{2} + \frac{1}{2} = 1$ (7 pieces total)